Chapter Review

Key Terms

solution of an equation A solution of an equation is a value of a variable that makes a true statement when substituted into the equation.

Key Concepts

8.1 Solve Equations Using the Subtraction and Addition Properties of Equality

- **Determine whether a number is a solution to an equation.**
  Step 1. Substitute the number for the variable in the equation.
  Step 2. Simplify the expressions on both sides of the equation.
  Step 3. Determine whether the resulting equation is true.

  If it is true, the number is a solution.
  If it is not true, the number is not a solution.
- **Subtraction and Addition Properties of Equality**
  - **Subtraction Property of Equality**
    For all real numbers *a*, *b*, and *c*,
    if $a = b$ then $a - c = b - c$.
  - **Addition Property of Equality**
    For all real numbers *a*, *b*, and *c*,
    if $a = b$ then $a + c = b + c$.
- **Translate a word sentence to an algebraic equation.**
  Step 1. Locate the "equals" word(s). Translate to an equal sign.
  Step 2. Translate the words to the left of the "equals" word(s) into an algebraic expression.
  Step 3. Translate the words to the right of the "equals" word(s) into an algebraic expression.
- **Problem-solving strategy**
  Step 1. Read the problem. Make sure you understand all the words and ideas.
  Step 2. Identify what you are looking for.
  Step 3. Name what you are looking for. Choose a variable to represent that quantity.
  Step 4. Translate into an equation. It may be helpful to restate the problem in one sentence with all the important information. Then, translate the English sentence into an algebra equation.
  Step 5. Solve the equation using good algebra techniques.
  Step 6. Check the answer in the problem and make sure it makes sense.
  Step 7. Answer the question with a complete sentence.

8.2 Solve Equations Using the Division and Multiplication Properties of Equality

- **Division and Multiplication Properties of Equality**
  - **Division Property of Equality:** For all real numbers *a*, *b*, *c*, and $c \neq 0$, if $a = b$, then $\frac{a}{c} = \frac{b}{c}$.
  - **Multiplication Property of Equality:** For all real numbers *a*, *b*, *c*, if $a = b$, then $ac = bc$.

8.3 Solve Equations with Variables and Constants on Both Sides

- **Solve an equation with variables and constants on both sides**
  Step 1. Choose one side to be the variable side and then the other will be the constant side.
  Step 2. Collect the variable terms to the variable side, using the Addition or Subtraction Property of Equality.
  Step 3. Collect the constants to the other side, using the Addition or Subtraction Property of Equality.
  Step 4. Make the coefficient of the variable 1, using the Multiplication or Division Property of Equality.
  Step 5. Check the solution by substituting into the original equation.
- **General strategy for solving linear equations**
  Step 1. Simplify each side of the equation as much as possible. Use the Distributive Property to remove any parentheses. Combine like terms.
  Step 2. Collect all the variable terms to one side of the equation. Use the Addition or Subtraction Property of Equality.
  Step 3. Collect all the constant terms to the other side of the equation. Use the Addition or Subtraction Property of Equality.
  Step 4. Make the coefficient of the variable term to equal to 1. Use the Multiplication or Division Property of Equality. State the solution to the equation.